

Qirui Mi

📍 Singapore ✉️ miqirui2021@ia.ac.cn ☎️ +86 16619789431

🌐 <https://miracle1207.github.io/miqirui.github.io/> in Qirui Mi 🌐 Miracle1207

Research Interests

AI: Reinforcement Learning (RL) agents, Large Language Model (LLM) agents, Multi-Agent Systems, Population-agent system, Game Theory.

AI for Economics: Agent-based modeling, Economic simulations, and Social simulations.

Education

Institute of Automation, Chinese Academy of Sciences, Beijing, China Aug 2021 – Present
 PhD Student in Pattern Recognition and Intelligent Systems
 Advisors: Prof. Jun Wang, Prof. Haifeng Zhang
Nanyang Technological University, Singapore Oct 2024 – Present
 Visiting PhD Student in Computer Science

Publications

EconGym: A Scalable AI Testbed with Diverse Economic Tasks June 2025
 Qirui Mi, Qipeng Yang, Zijun Fan, Wentian Fan, Heyang Ma, Chengdong Ma, Siyu Xia, Bo An, Jun Wang, Haifeng Zhang
Submitted to NeurIPS 2025. [arXiv:2506.12110](#)

MF-LLM: Simulating Population Decision Dynamics via a Mean-Field Large Language Model Framework Apr 2025
 Qirui Mi, Mengyue Yang, Xiangning Yu, Zhiyu Zhao, Cheng Deng, Bo An, Haifeng Zhang, Xu Chen, Jun Wang
Submitted to NeurIPS 2025. [arXiv:2504.21582](#)

Learning Macroeconomic Policies through Dynamic Stackelberg Mean-Field Games Apr 2025
 Qirui Mi, Zhiyu Zhao, Chengdong Ma, Siyu Xia, Yan Song, Mengyue Yang, Jun Wang, Haifeng Zhang
ECAI 2025. [arXiv:2403.12093](#)

Mean Field Correlated Imitation Learning Apr 2024
 Zhiyu Zhao, Chengdong Ma, Qirui Mi, Ning Yang, Xue Yan, Mengyue Yang, Haifeng Zhang, Jun Wang, Yaodong Yang
AAMAS 2025.

Large Language Models Play StarCraft II: Benchmarks and a Chain of Summarization Approach Dec 2024
 Weiyu Ma, Qirui Mi, Yongcheng Zeng, Xue Yan, Runji Lin, Yuqiao Wu, Jun Wang, Haifeng Zhang
NeurIPS 2024.

TaxAI: A Dynamic Economic Simulator and Benchmark for Multi-Agent Reinforcement Learning Sept 2023
 Qirui Mi, Siyu Xia, Yan Song, Haifeng Zhang, Shenghao Zhu, Jun Wang
AAMAS 2024.

Joint Caching and Transmission in the Mobile Edge Network: A Multi-Agent Learning Approach Dec 2021
 Qirui Mi, Ning Yang, Haifeng Zhang, Haijun Zhang, Jun Wang
IEEE GLOBECOM 2021.

Talks

2025 Organizing and presenting the AAMAS 2025 Computational Economic Competition. *AAMAS 2025*, Detroit, USA.

2025 Mean Field Correlated Imitation Learning. Oral presentation at *AAMAS 2025*, Detroit, USA.

2025 Game-Theoretic Agents for Economic and Social Governance. *Dadao Forum*, Shandong Institute of Business and Technology.

2024 Game-Theoretic Agents for Economic Problems. Invited talks at *Shanghai Jiao Tong University* (JHC Lecture Series) and *Fudan University*.

2024 TaxAI: A Dynamic Economic Simulator and Benchmark for Multi-Agent Reinforcement Learning. Invited talk at *DAI 2024*, Singapore.

2024 TaxAI: A Dynamic Economic Simulator and Benchmark for Multi-Agent Reinforcement Learning. Oral presentation at *AAMAS 2024*, Auckland.

Services

Main organizer of the AAMAS 2025 Computational Economics Competition: Optimal Tax Policy with Heterogeneous Agent Models. [Website](#) 

Main organizer of the AAMAS 2024 Computational Economics Competition: Optimal Tax Policy. [Website](#) 

Main organizer of the CCF Computational Economics Competition. [Website](#) 

Reviewers: Neurips, AAAI, ICLR, AAMAS, ICML, IJCAI.

Honors

Outstanding Student Leader	2024
Outstanding Merit Student	2023
Outstanding Graduate of Beijing Colleges and Universities	2021
Outstanding Undergraduate Thesis of Beijing Colleges and Universities	2021
Meritorious Winner, MCM (USA)	2019
Second Prize, 10th ‘Challenge Cup’ for Beijing College Students	2019